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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0915- JAVA PROGRAMMING

Development of a Java-Based Computer Recommendation and Performance Analysis System

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Project Overview

Bridging Workloads with Optimal Hardware

Selecting an optimal computer configuration requires matching complex user applications with rapidly changing hardware architectures.

1. Workload Ingestion: Captures user budget, CPU usage, RAM, and storage demands.

2. Hardware Mapping: Expert decision engine recommends CPU, GPU, and memory setups.

3. Performance Analytics: Calculates execution time, power usage, and PDF summaries.

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Problem Statement

Static Pre-builds

Generic PC recommendation engines rely on rigid pre-configured lists that fail to adapt to dynamic workload demands.

Resource Mismatch

Users misallocate budget by overspending on unnecessary specs while underpowering critical application bottlenecks.

No Analytics

Traditional tools recommend hardware without offering execution time estimation, TDP power calculations, or reports.

Key Objectives

- **Requirement Profiling:** Collect budget, workload class, CPU, RAM, and storage parameters.
- **Rule-Based Recommendation:** Build an expert matrix to recommend optimal CPU cores, cache, RAM, and GPU.
- **Performance Evaluation:** Calculate execution time estimates, TDP wattage, and overall architecture score.
- **Automated Reporting:** Export interactive charts and downloadable PDF diagnostic summaries.

Rule Engine Matrix

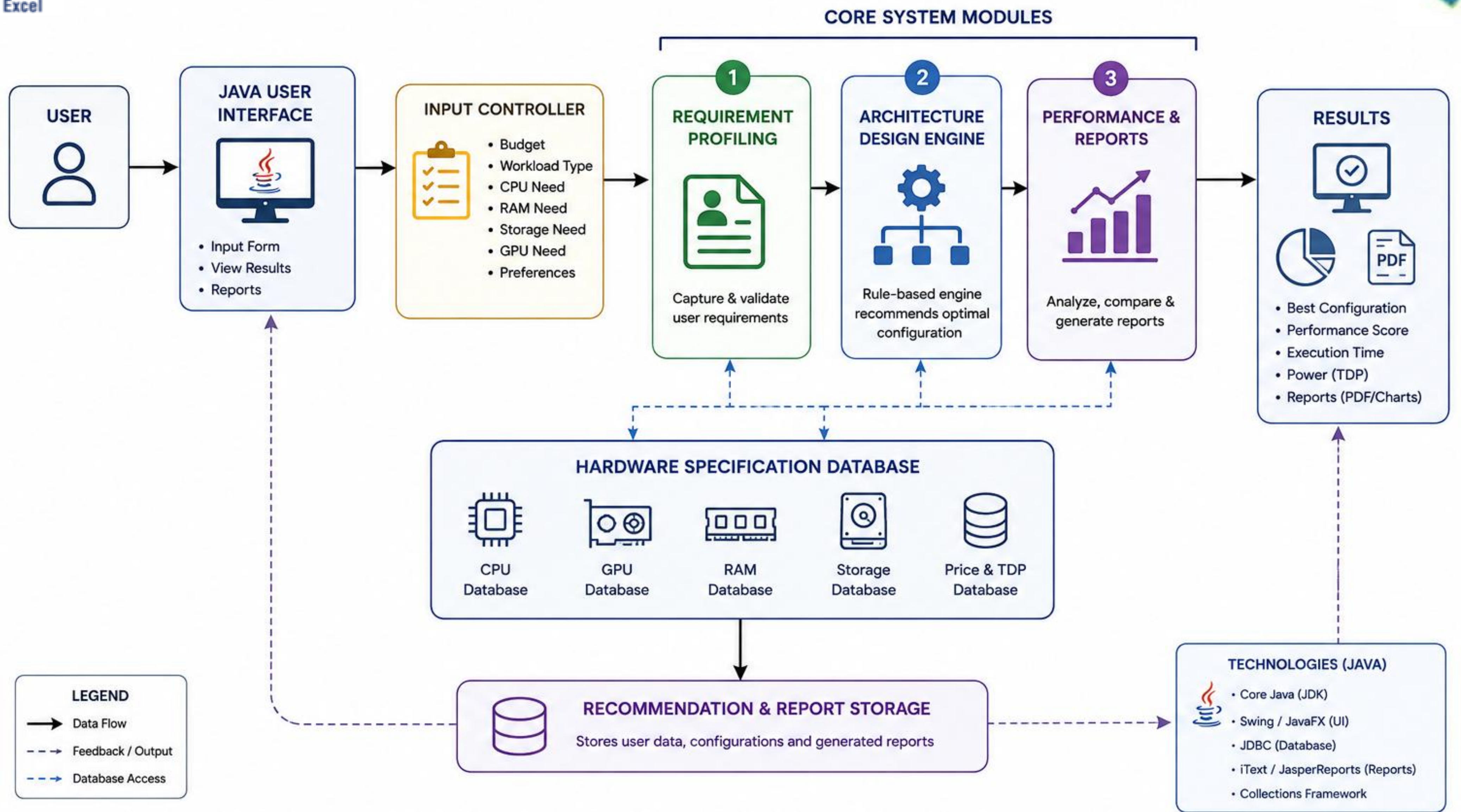
100%

Automated mapping of complex user parameters into concrete, non-bottlenecked system hardware configurations.

Literature Survey

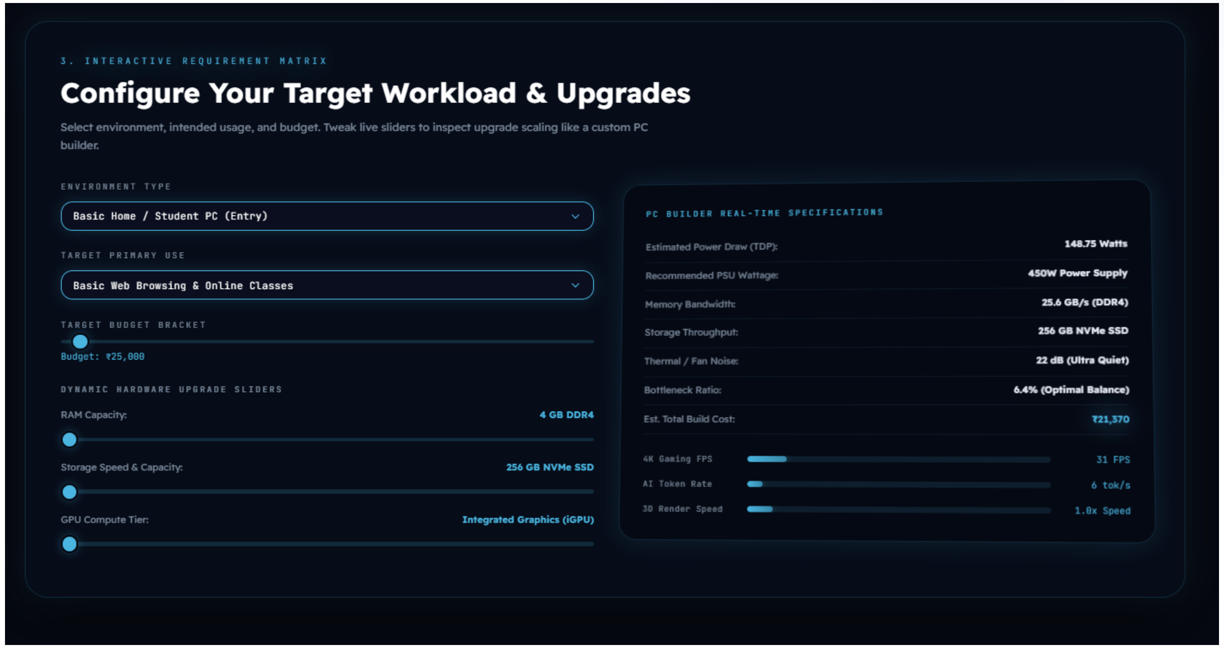
Paper / Project Name	Authors	Methodology Used	Key Advantages	Disadvantages & Limitations
PC Component Recommendation System.	Mishra et al.	KNN Collaborative Filtering	Finds popular component pairings from user trends.	Fails on new hardware without rating history.
Expert PC Advisory System.	Abusua'da & Hedaya	Fuzzy Rule-Based Inference	Guarantees 100% hardware physical compatibility.	Needs manual updates for every new release.
Knowledge-Based Hardware Configurator.	Gupta & Patel	Multi-Objective Genetic Algorithm	Optimizes raw hardware specs within budget caps.	High computation time; lacks execution metrics.
PCPartPicker Configurator.	Commercial Platforms	Static Relational DB Queries	Shows real-time market prices and stock levels.	Lacks workload profiling and TDP analytics.

System Architecture Framework



Module 1: User Requirement Collection

Collect User Information: Gather basic user details such as name, profession, and experience level.
Identify User Needs: Determine the primary purpose of the computer, such as programming, gaming, or office work.
Collect Budget: Record the user's preferred budget range for selecting suitable computer systems.
Analyze Performance Requirements: Identify the required processor, RAM, storage, and graphics specifications.
Gather User Preferences: Collect preferences for brand, operating system, laptop/desktop, and other features.
Validate and Store Data: Verify the entered information and store it securely for generating recommendations.



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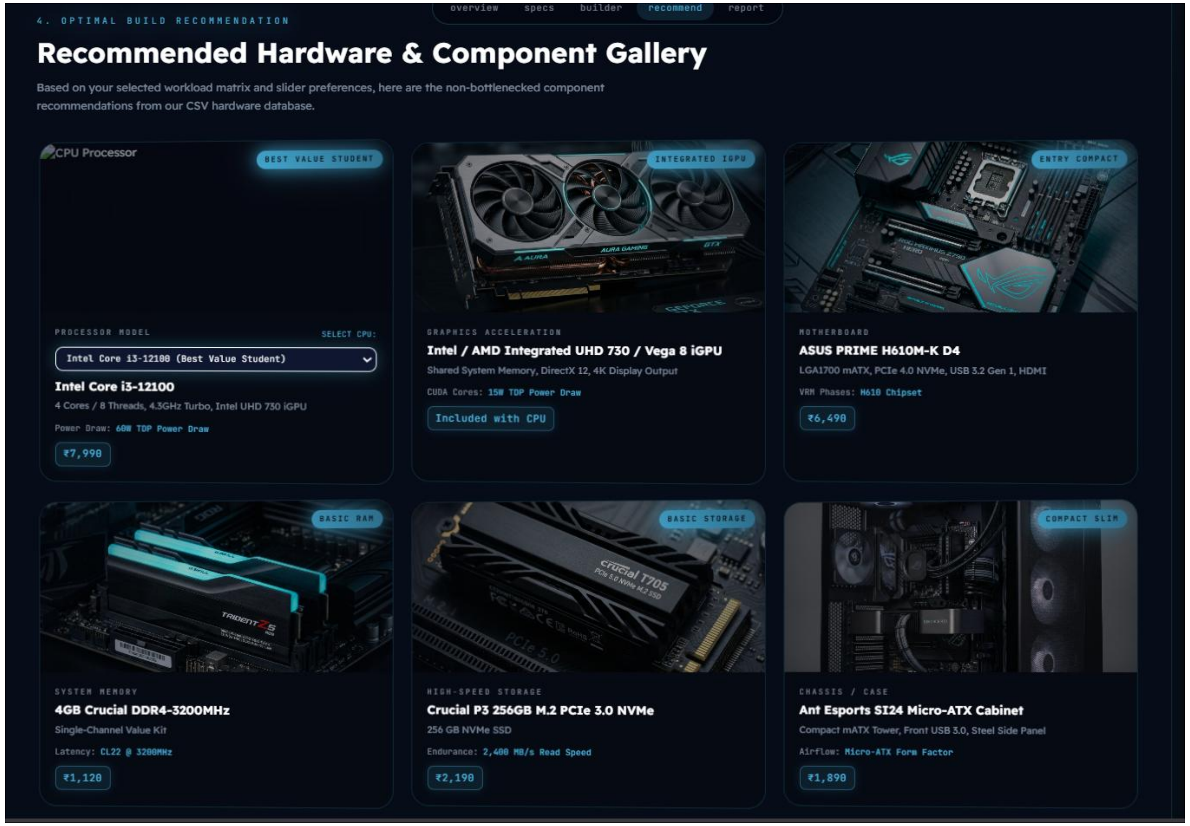
Module 2: Architecture Design Engine

System Architecture Design: Define the overall structure and workflow of the recommendation platform.
Component Integration: Connect the user interface, database, recommendation engine, and analysis modules.
Recommendation Logic Design: Develop the logic for matching user requirements with suitable computer systems.
Database Architecture: Organize and manage computer specifications, user profiles, and recommendation data.
Performance Analysis Framework: Design methods to evaluate and compare the performance of recommended systems.
System Optimization: Improve the architecture for better scalability, efficiency, and faster recommendation generation.



Module 3: Performance & Report Module

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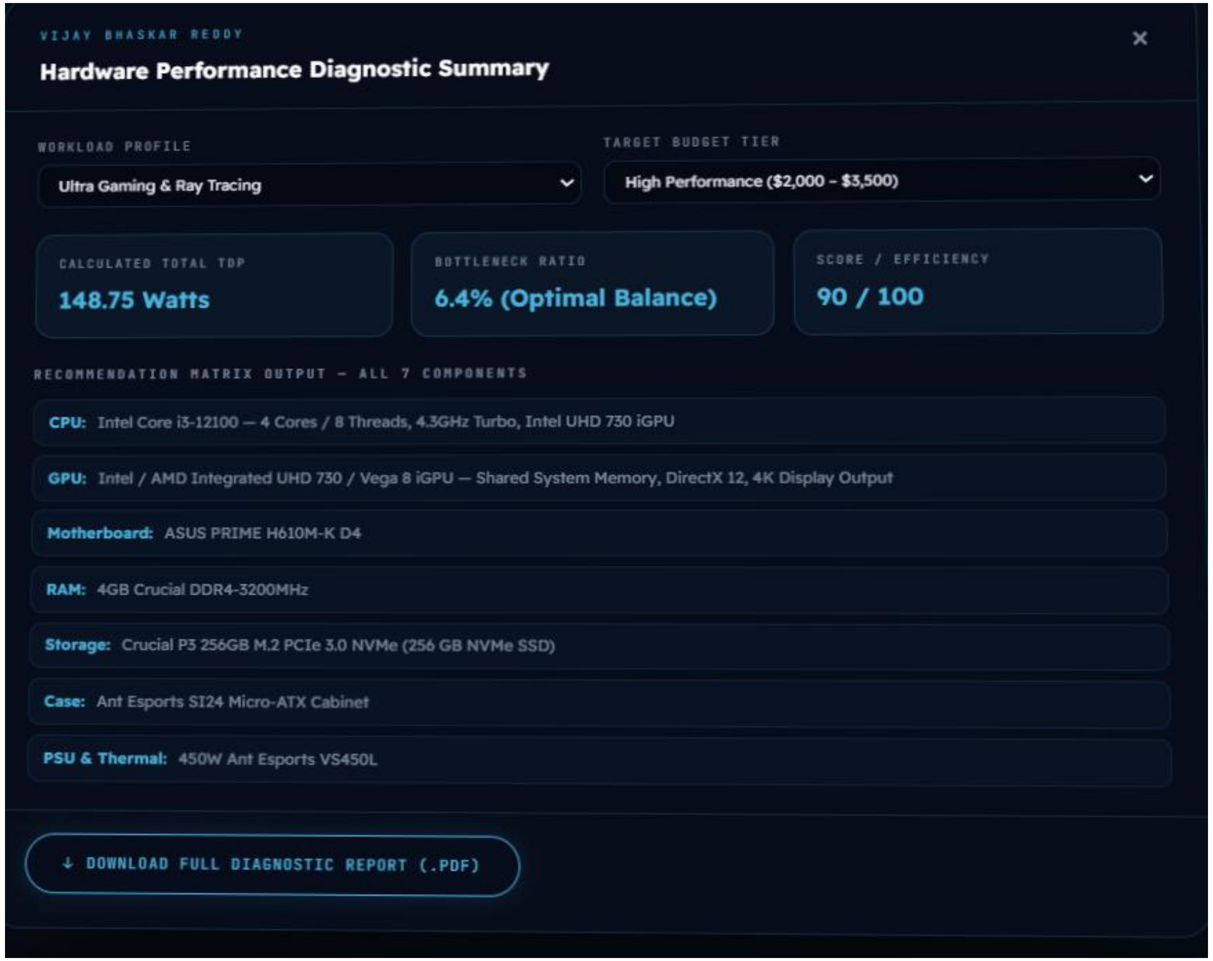
Performance Evaluation: Analyze the performance of recommended computer systems based on user requirements.
Specification Comparison: Compare multiple computer systems using key hardware and software specifications.
Report Generation: Generate detailed reports with recommendations and performance analysis results.
Performance Visualization: Display results using charts, graphs, and comparison tables for better understanding.
Recommendation Summary: Provide the best computer recommendation with reasons for selection.
Report Storage and Export: Save reports and allow users to export them in formats such as PDF or Excel.



Hardware Performance Diagnostic Report

COMPUTER RECOMMENDATION & PERFORMANCE ANALYSIS SYSTEM — VIJAY BHASKAR REDDY

Vijay Bhaskar Reddy
Generated: 23 August 2026 at 2:19 pm
Budget: ₹25,000
Environment: BASIC

BUILD CONFIGURATION

ENVIRONMENT	PRIMARY USE	BUDGET TARGET
Basic	**BASIC_WEB**	**₹25,000**
RAM CAPACITY	STORAGE	GPU TIER
4 GB DDR4	**256 GB NVMe SSD**	**Integrated Graphics (iGPU)**

PERFORMANCE METRICS

EST. TOTAL BUILD COST	ESTIMATED TDP	RECOMMENDED PSU	MEMORY BANDWIDTH
₹21,370	**148.75 Watts**	**450W Power Supply**	**25.6 GB/s (DDR4)**
STORAGE THROUGHPUT	THERMAL / FAN NOISE	BOTTLENECK RATIO	EFFICIENCY SCORE
256 GB NVMe SSD	**22 dB (Ultra Quiet)**	**6.4% (Optimal Balance)**	**90 / 100**

RECOMMENDED COMPONENTS — 7-COMPONENT BUILD

	COMPONENT	MODEL & SPECIFICATIONS	KEY STAT	BADGE	
🖥	Processor (CPU)	**Intel Core i3-12100** 4 Cores / 8 Threads, 4.3GHz Turbo, Intel UHD 730 iGPU	60W TDP Power Draw	BEST VALUE STUDENT	₹7,990
🎮	Graphics (GPU)	**Intel / AMD Integrated UHD 730 / Vega 8 iGPU** Shared System Memory, DirectX 12, 4K Display Output	15W TDP Power Draw	INTEGRATED IGPU	Included with CPU
🔌	Motherboard	**ASUS PRIME H610M-K D4** LGA1700 mATX, PCIe 4.0 NVMe, USB 3.2 Gen 1, HDMI	H610 Chipset	ENTRY COMPACT	₹6,490

Summary Table of System Modules

Module Name	Core Objective	Key Inputs / Specs	Output & Tech Stack
Module 1: Requirement Profiling	Collect user parameters & analyze workload needs	Budget, workload type, CPU, RAM, storage, app type	Structured requirement profile
Module 2: Architecture Builder	Generate optimal computer architecture configuration	CPU cores, clock speed, cache, RAM, storage, GPU	Optimized system configuration
Module 3: Performance Module	Evaluate configuration and generate analytics reports	Score, execution time, power usage (TDP), upgrade score	Interactive charts & PDF report

Key Advantages & Future Scope

System Advantages

- **Workload Efficiency:** Eliminates hardware mismatch and bottlenecks.
- **Cost Optimization:** Maximizes performance within budget limits.
- **Comprehensive Reports:** Downloadable PDF performance summaries.

Future Scope

- **Live Pricing APIs:** Fetch real-time hardware component prices.
- **Machine Learning:** Predictive models for workload profiling.
- **Hardware Validation:** Automated socket & motherboard VRM checks.

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THANK YOU


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Configure Your Target Workload & Upgrades
GPS Map Camera
Google
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